Machine Learning Engineer Roadmap & Tools Guide

1. Foundations

1. Mathematics: Linear Algebra, Probability & Statistics, Basic Calculus
2. Programming: Python (loops, functions, OOP, file handling, exceptions)

2. Core Libraries

1. NumPy – numerical operations
2. Pandas – data manipulation
3. Matplotlib / Seaborn – data visualization

3. Machine Learning Basics

1. Supervised Learning: Linear Regression, Logistic Regression, KNN, Decision Tree, Random Forest, SVM, Naive Bayes
2. Unsupervised Learning: K-Means, Hierarchical Clustering, PCA
3. Concepts: Overfitting, Underfitting, Cross Validation, Bias-Variance Tradeoff

4. Tools & Libraries

1. Scikit-learn (core ML library)
2. XGBoost / LightGBM (advanced models)
3. TensorFlow / PyTorch (deep learning)

5. Data Handling & Feature Engineering

1. Handling missing values
2. Encoding techniques (Label, One-hot)
3. Feature scaling (Normalization, Standardization)
4. Feature selection

6. Projects

1. Beginner: House Price Prediction, Spam Detection
2. Intermediate: Fake News Detection, Recommendation System
3. Advanced: Chatbot, Image Classifier

7. Deployment

1. Flask / FastAPI
2. Streamlit
3. REST APIs

8. Version Control & Databases

1. Git & GitHub
2. SQL (SELECT, JOIN, GROUP BY)
3. MongoDB (basic)

9. Advanced Tools (MLOps & Cloud)

1. Docker, Kubernetes
2. MLflow, Airflow
3. AWS / Google Cloud / Azure

Final Skill Stack

1. Python, NumPy, Pandas
2. Scikit-learn / PyTorch
3. SQL, GitHub
4. Deployment (Flask/Streamlit)
5. Projects on GitHub